

EVE ECC — Avatar Pipeline Research Report

Sep 21, 2026

1. Where we are

Repository: `tyronne-os/eve-ecc`, branch `main`, last commit `13167ce` (Aug 8, 2026, 23:50 CDT) — *"docs: the orchestration pivot — read before touching the avatar pipeline."* Working tree clean, no open PRs or issues. The project went quiet for roughly six weeks after that commit; a DNS man-in-the-middle attack on the user's machine forced a full local wipe. On resuming (Sept 21), repo integrity was verified directly against GitHub — local HEAD matches remote exactly, `git fsck` clean, no secrets in any commit history, `.env` never tracked. The repo itself was never touched by the attack; only the local machine was.

The state the last commit left us in. The team had built a CSS-based "life layer" on EVE's static portrait — parallax depth, a breath curve, hair lag, a masked eyelid sweep — and it scored 96/100 on the project's own Realness Index. The director looked at it and said: *"It is not alive — this is a blinking portrait."* He was right, and the pivot doc names two failures rather than softening them:

- **Cosmetics mistaken for science.** Transform math on a flat photograph is theater. Real motion from a 2D image requires a model that actually drives facial geometry — diffusion-based portrait animation — not CSS pretending to.
- **A high score on the wrong metric,** called the worse failure of the two. The certification's MOTION pillar read transform *values* instead of what was actually painted on screen, and passed dead output twice — once while depth layers animated perfectly *behind an opaque portrait*, 100% correct and 100% invisible.

The reframe that came out of that night: *"Assembling and connecting, when it comes to building a human avatar, are unrelated problems."* Every component already existed — NIM endpoints, Nemotron, a live cortex, a persona, a phoneme pipeline — but nothing owned making them behave as one living thing. The original plan to fix this was purely architectural: adopt Microsoft Agent Framework (Magnetic as supervisor, Handoff for node-to-node signal-path conversation, Group Chat for diagnosis quorum) on top of the existing `services/ace-controller`, which already exposes `/v1/nodes/{id}/converse`, `/v1/speak`, and `/v1/report`.

Where this research extends that plan. Re-opening the project surfaced a second, equally real gap sitting underneath the orchestration one: the render pipeline itself — Audio2Face-3D → AnimGraph → Omniverse Stream — is a rigged-mesh pipeline that was never actually reachable (hosted Maxine endpoints 404'd) and was the wrong tool for a 2D photograph regardless. So the research below runs in parallel to the orchestration work: not a replacement for the Magentic/Handoff plan, but the missing answer to what the render node in that graph should actually *be*.

2. What we're researching, and why

The brief has four hard constraints, and each one eliminated a large share of what's out there:

- 1. Cost ceiling: ~\$0.40/hr, GPU-resident, always warm.** Not "cheap to run once" — the model has to stay loaded and hot so the avatar is alive the instant the page loads, with headroom to spare rather than compute maxed out just reaching real-time. This ruled out almost every model trained at frontier scale (14B–22B parameters, 5–8 H800/H200 GPUs) regardless of how good its numbers were.
- 2. No mesh, no rig, no gaming pipeline.** The ask is a single 2D photograph in, 3D-aware motion out — real head rotation, parallax, occlusion — not a literal reconstructed 3D asset. This is a deliberate rejection of the Audio2Face-3D-style pipeline already sitting in the repo's node graph, and it also ruled out the explicit-3D-reconstruction papers found during research (they solve a different problem, at a different speed, discussed in section 3).
- 3. Maximum realism and conversational awareness, live.** Not just speaking — *listening* has to read as alive too, which is a much rarer property to find shipped in open work than lip-sync is.
- 4. Build on successful open-source work with integrity, without acquiring liability.** The user, running what amounts to a stealth lab with full NVIDIA Enterprise/NGC backing, was explicit about wanting to credit and build on real prior work rather than start from zero or attempt to reproduce a paper cold — and wanted the chosen foundation permissive enough that further open-source components could be merged into it later without one of them quietly attaching a copyleft or field-of-use obligation to the whole product. That turned "pick a license" into "audit the dependency tree," which is where several models that looked clean at the top level turned out not to be (section 3).

A fifth constraint shaped the search without being stated outright: the user already holds NVIDIA Enterprise and NGC access, which is not nothing — NVIDIA's Audio2Face-2D ("Speech Live Portrait") NIM does exactly the photo-plus-audio job being searched for, and turned out to be sitting in the user's own entitlements the whole time, unreachable in the original attempt only because the earlier team tried to hit hosted Maxine endpoints that 404'd rather than pulling the NGC container directly.

3. The shortlist

Every entry below was checked against source — cloned from GitHub and read directly where possible, since this session's network policy blocks huggingface.co and arxiv.org (an allowlist gap, unrelated to the earlier attack, not yet corrected). License claims are LICENSE-file-verified unless noted; hardware claims are README-stated unless noted as code-derived.

Project	Date	License	Hardware for real-time	Verdict
SoulX-FlashHead-Lite (1.3B)	Feb 2026	Apache-2.0, clean dependency tree (verified)	1× RTX 4090 @ 96 FPS, or 3 concurrent streams @ 25+ FPS	Kept — foundation
LeapTalk (LoRA on FlashHead)	Jul 2026	Apache-2.0, clean dependency tree (verified)	Same base; 1 NFE, up to 200 FPS claimed in "Lite" setting (unverified on real hardware)	Kept — render/UI layer
SoulX-Duplug (0.6B)	Mar-Jul 2026	Apache-2.0	CPU-light semantic VAD, pairs with FlashHead family	Kept — turn-taking/listening state
NVIDIA Audio2Face-2D (NIM)	—	NVIDIA AI Enterprise SLA + AI Foundation Models Community License (not open source; already entitled)	Explicitly targets Ada/Ampere/Turing/Blackwell — excludes A100/H100	Kept — quality reference, run in parallel
LipForcing (14B / 1.3B)	Jun-Jul 2026	Apache-2.0	14B: ~37–50GB VRAM (fits L40S 48GB); sub-millisecond time-to-first-frame	Reference architecture — too heavy as primary base

Project	Date	License	Hardware for real-time	Verdict
Hallo-Live	Apr 2026	MIT, clean	2× H200 @ 20.4 FPS, 0.94s latency	Reference architecture only — over budget
Avatar-Forever	Aug 2026	Repo LICENSE present, but depends on Gemma-3	1× H100 @ 27.2 FPS, 768×512	Ruled out — Gemma Terms of Use is not OSI-approved and requires passing its use-restrictions through to end users
daVinci-MagiHuman (15B)	2026	Apache-2.0 at root, but depends on Stable Audio	1× H100, offline clips only (no streaming)	Ruled out as base — Stability AI Community License is free only under \$1M revenue; SR modules (540p/1080p) still worth adopting standalone
AVTR-1	2026	Split licensing: weights community-licensed, renderer + streamer noncommercial-only regardless of revenue, face detector (InsightFace) research-only	L40 @ 84ms/chunk (2.4× real-time)	Ruled out — the two components that make it "live" are the ones not licensed for commercial use
LiveAvatar (Alibaba-Quark)	Dec 2025	Apache-2.0	5× H800 80GB (single-GPU needs 80GB+)	Ruled out — far over budget
OmniForcing	Mar 2026	LTX-2 Community License (non-standard)	8× H200 minimum	Ruled out — license + hardware

Project	Date	License	Hardware for real-time	Verdict
Omni-LiveAvatar	Aug 2026	—	Claims 33× speedup on 1× H200	Ruled out — vaporware. README states plainly: <i>"Code and checkpoints are not released yet."</i>
SplatShot	Jun 2026	No LICENSE file	1× A100, 24GB, 10–15 min per photo , static output only (doesn't talk)	Ruled out — fails "brief processing" by two orders of magnitude, and isn't audio-driven
EmoTaG	2026 (CVPR'26)	Research-use only	Few-shot, 5-second personalization clip	Ruled out — license
Wan-Streamer (Alibaba)	Jun–Jul 2026	Apache-2.0 license file present, no checkpoint released ; API-only beta since Aug 2026	~550ms full-duplex latency (single unified model, no fixed rig)	Not usable today — the clearest north star for <i>architecture</i> , unusable for <i>code</i>
Vidu S2-Avatar	Sep 15, 2026	Credit-metered API only	720p, 25–42 FPS	Not open — reference for what the commercial ceiling looks like

4. The direction, and why

Recommended stack:

- **SoulX-FlashHead-1.3B** as the render foundation — chosen over every larger, flashier model specifically because its dependency tree is clean all the way down, not just at the repo root. ~20 dependencies checked (opencv, diffusers, transformers, accelerate, gradio, xformers, mediapipe, librosa) — all Apache-2.0/BSD/MIT, nothing gated, nothing research-restricted. It comfortably clears the cost ceiling with room to spare: 96 FPS against a 25 FPS real-time requirement on a single RTX 4090, which is exactly the kind of headroom "alive upon load" needs — spare compute burning on idle presence instead of a cold start.

- **LeapTalk** as the drop-in render/UI layer on the *same* base — its own ~160-package dependency tree audited the same way, equally clean. It matters beyond raw speed: its shipped web demo already implements almost exactly the interface being designed — upload a portrait, brief processing, streaming talking video on the left, chat/input on the right. Someone already did the “learn from it, build on it” step this research was asked to find, and published it Apache-2.0.
- **SoulX-Duplug** (0.6B, Apache-2.0) for the listening/turn-taking layer — it classifies audio into idle/nonidle/speak/blank states in real time, which is the missing half of “conversational awareness” that almost nothing else in the entire search actually modeled (the one model that did, AVTR-1, turned out to be the one with the licensing problem).
- **NVIDIA Audio2Face-2D**, run in parallel as the quality ceiling the open pipeline is measured against — not open source, but already covered by the user’s existing NGC/Enterprise entitlement, and engineered for exactly the consumer/prosumer cards (Ada/Ampere/Turing) this budget targets rather than the datacenter cards it explicitly excludes.

Why this over the alternatives: every heavier or more “complete”-looking option in section 3 failed one of the four constraints — too expensive (LiveAvatar, OmniForcing, Hallo-Live), not actually released (Omni-LiveAvatar), too slow per-photo and not audio-driven (SplatShot), or clean at the surface but carrying a restrictive dependency underneath (Avatar-Forever’s Gemma-3, daVinci-MagiHuman’s Stable Audio, AVTR-1’s InsightFace and noncommercial renderer). The FlashHead/LeapTalk/Duplug stack is the only one that is simultaneously cheap enough, fast enough, and clean enough at every layer to build a product on without a license audit surfacing a problem after the fact.

On building on someone else’s open-source work with integrity: this is standard, well-trodden practice, not something to obscure. The concrete mechanism agreed on:

1. A `NOTICE` file at the repo root, one entry per upstream project (FlashHead, LeapTalk, Duplug, wav2vec2) — name, source URL, license, and what was changed. This is what Apache-2.0 Section 4 actually requires, not just a suggestion.
2. A CI dependency license scan (`pip-licenses` against `requirements.txt`) on every merge, gating anything that isn’t OSI-approved permissive — this is precisely the check that would have caught the Gemma-3 dependency automatically, before a human had to notice it by hand.
3. Never use an upstream project’s name or branding to imply their endorsement — standard across MIT and Apache-2.0 alike.

4. Apache-2.0 already disclaims warranties and caps liability (Sections 7–8) for both the original author and, by inheritance, anyone building on it — that protection comes with the license and doesn't need to be constructed, only not stripped out on redistribution.

This practice is what makes "merge other open source without flags" achievable: a foundation that is permissive at every layer, not just its own LICENSE file, is what actually allows other permissively-licensed pieces to be merged in later without one of them quietly attaching a copyleft or revenue-gated obligation to the whole product.

5. Open questions and next steps

- **LeapTalk FPS resolved: working number is 96 FPS on 1x RTX 4090, base-verified.** FlashHead-1.3B delivers 96 FPS on one RTX 4090 (README-verified). LeapTalk runs 1 NFE on the same base, setting its floor at 96 FPS. The unverified Lite-setting claim of ~200 FPS is dropped from all pipeline documents until measured on hardware. Working assumption for the pipeline: **96 FPS / 10.4ms per frame** -- 4x headroom above the 25 FPS real-time bar, enough to absorb the daVinci SR post-pass without touching the budget. Note: the UPON LOAD 3.77s failure is cold-start / model-initialization latency, not steady-state FPS -- different instrument, different fix, tracked separately.
- **VRAM footprint is bounded, not measured.** From the code (bf16 weights, no offload flags in the single-GPU path, three concurrent streams claimed on one 24GB card), per-stream footprint has to be comfortably under ~7GB — but that's a derived bound, not a profiled number.
- **The listening-reactive motion signal still needs to be built.** AVTR-1 proved the concept (a separate audio track driving reactive idle motion, not just speech) but its implementation isn't licensed for commercial use. The plan is to reimplement the *idea* — wiring Duplug's idle/nonidle/speak/blank state output into FlashHead/LeapTalk's render loop — rather than import restricted code.
- **Resolution ceiling.** Current real-time open models top out around 512×512–768×512. The realistic path past that is a separate super-resolution pass after generation, not a bigger base model — daVinci-MagiHuman ships exactly that as standalone Apache-2.0 540p/1080p modules, worth adopting even though its base generator was ruled out.
- **Where Audio2Face-2D sits in the pipeline** — as an offline quality benchmark to validate the open stack against, or as a selectable node inside the ACE Cortex graph itself, still an open design call.

- **Immediate next step, already queued as an urgent task:** stand up a new repository seeded with FlashHead + LeapTalk + Duplug, the NOTICE file, and the CI license scanner — blocked only on the user being back at a laptop.
- **Offered and not yet taken:** running LeapTalk's streaming demo against a real photo on real hardware to get the actual first-frame latency and FPS numbers this report currently has to describe as claims rather than measurements.

6. Update — 2026-09-21

Infra status: Kiro Web sandbox fully provisioned — GitHub connected (native OAuth), and four secrets loaded globally under Agent settings > Sandbox > Secrets: Hugging Face token, Google Cloud Compute OAuth client, NVIDIA NIM key, NVIDIA Enterprise credential. No longer blocked on "being back at a laptop" — build proceeds from Kiro Web.

Work Order 1 (issued today): clone SoulX-FlashHead-1.3B, LeapTalk, and SoulX-Duplug into `services/render-pipeline/`, add a NOTICE file, provision/reuse a GCP GPU instance, run a smoke test (one photo + one audio clip) and report measured FPS, first-frame latency, and VRAM — replacing the report's prior estimates rather than confirming them by assumption. On success, deploy the working service to a Hugging Face Space, and add a CI license scan (`pip-licenses` against `requirements.txt`) gating non-permissive dependencies. Status: in progress, numbers pending.

Correction: an earlier note in this thread about switching "the model" to Opus 5 referred to the chat model answering questions in this research thread, not a component of the eve-ecc stack. No pipeline model choice changed as a result.